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Point values are assigned for each question.

Points earned: ____ / 100, = ____ %

1. Find an upper bound for $f(n) = n^4 + 10n^2 + 5$. Write your answer here: $O(n^4)$ (4 points)

Prove your answer by giving values for the constants c and n_0 . Choose the smallest integral value possible for c . (4 points)

$$c = 2, n_0 = 4$$

2. Find an asymptotically tight bound for $f(n) = 3n^3 - 2n$. Write your answer here: $O(n^3), \Omega(n)$ (4 points)

Prove your answer by giving values for the constants c_1, c_2 , and n_0 . Choose the tightest integral values possible for c_1 and c_2 . (6 points)

$$c_1 = 3, c_2 = 2, n_0 = 2$$

3. *Is $3n - 4 \in \Omega(n^2)$? Circle your answer: yes / **no**. (2 points) **NO**

If yes, prove your answer by giving values for the constants c and n_0 . Choose the smallest integral value possible for c . If no, derive a contradiction. (4 points)

When proving the possibility of this claim being true, it was clear that it was not the case. As shown in the work below (end of this document), with the c value of 1 and n being 2, the inequality disproves this. Since there are values that disprove the inequality $3n - 4 \in \Omega(n^2)$ is false. $c = 1, n = 2$, so this is a contradiction.

4. Write the following asymptotic efficiency classes in **increasing** order of magnitude.

$O(n^2), O(2^n), O(1), O(n \lg \lg n), O(n), O(n!), O(n^3), O(\lg \lg n), O(n^n), O(n^2 \lg \lg n)$ (2 points each)

$$O(1), O(\lg n), O(n), O(n \lg n), O(n^2), O(n^2 \lg n), O(n^3), O(2^n), O(n!), O(n^n)$$

5. Determine the largest size n of a problem that can be solved in time t , assuming that the algorithm takes $f(n)$ milliseconds. Write your answer for n as an integer. (2 points each)

a. $f(n) = n, t = 1 \text{ second}$ $n=1000$

b. $f(n) = n \lg n, t = 1 \text{ hour}$ $n=204,094$

c. $f(n) = n^2$, $t = 1$ hour $n=1897$

d. $f(n) = n^3$, $t = 1$ day $n=442$

e. $f(n) = n!$, $t = 1$ minute $n=9$

6. Suppose we are comparing two sorting algorithms and that for all inputs of size n the first algorithm runs in $4n^3$ seconds, while the second algorithm runs in $64n \lg n$ seconds. For which integral values of n does the first algorithm beat the second algorithm? 6 (4 points)

Explain how you got your answer or paste code that solves the problem (2 point):

$4n^3 \geq 64n \lg n$, and this is only true when n is greater than or equal to 6, so the smallest value has to be 6.

7. Give the complexity of the following methods. Choose the most appropriate notation from among O , Θ , and Ω . (8 points each)

```
int function1(int n) {
    int count = 0;
    for (int i = n / 2; i <= n; i++) {
        for (int j = 1; j <= n; j *= 2) {
            count++;
        }
    }
    return count;
}
```

Answer: $O(n \lg n)$

```
int function2(int n) {
    int count = 0;
    for (int i = 1; i * i * i <= n; i++) {
        count++;
    }
    return count;
}
```

Answer: $O(\sqrt[3]{n})$

```
int function3(int n) {
    int count = 0;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            for (int k = 1; k <= n; k++) {
                count++;
            }
        }
    }
    return count;
}
```

Answer: $O(n^3)$

```
int function4(int n) {  
    int count = 0;  
    for (int i = 1; i <= n; i++) {  
        for (int j = 1; j <= n; j++) {  
            count++;  
            break;  
        }  
    }  
    return count;  
}
```

Answer: $O(n)$

```
int function5(int n) {  
    int count = 0;  
    for (int i = 1; i <= n; i++) {  
        count++;  
    }  
    for (int j = 1; j <= n; j++) {  
        count++;  
    }  
    return count;  
}
```

Answer: $O(n)$

$$1. n^4 + 10n^2 + 5 \leq 2n^4 \quad \begin{matrix} n=4 \\ c=2 \end{matrix}$$

$$0 \leq n^4 - 10n^2 - 5$$

$$0 = n^4 - 10n^2 - 5$$

$$n = -3.237, n = 3.237 \rightarrow 4$$

$$2. 3n^3 - 2n \geq 2n^3 \quad \begin{matrix} c_1 = 2 \\ n_0 = 2 \end{matrix} \quad 3n^3 - 2n \leq 3n^3 \quad \begin{matrix} c_2 = 3 \\ n_0 = 2 \end{matrix}$$

$$n^3 - 2n \geq 0$$

$$n(n^2 - 2) \geq 0$$

$$(n + \sqrt{2})(n - \sqrt{2}) \geq 0$$

$$n = \sqrt{2}$$

$$-2n \leq 0$$

$$n \geq 0$$

$$n = 1$$

The square root of 2 is greater than $n=1$, so both have to be rounded up to keep both true.

$$3. 3n - 4 \geq n^2 \quad c=1$$

$$3n \geq n^2 + 4$$

There is no value ^{for n} in which this is true.

$$0 \geq n^2 - 3n + 4$$

$$0 \geq (n-4)(n+1)$$

$$n \leq 4 \quad n \leq -1$$

$$n=2$$

$$3(2) - 4 \geq 1 \cdot 2^2$$

$$2 \geq 4$$

I pledge my honor that I have abided by the Stevens Honor System. Octavio Morales